Energy-Efficient and Near Real-Time Routing Algorithm for Disaster Monitoring in Wireless Sensor Networks

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Abstract

Abstract— Wireless Sensor Networks (WSNs) deployed in disaster-monitoring scenarios require energy-efficient, low-latency multi-hop routing under strict resource constraints. Existing multi-agent reinforcement learning (MARL) approaches such as QMIX and QTRAN treat the routing problem independently, ignoring the network topology and suffering from memory explosion at scale. We propose **Graph-Attentive Policy Fusion (GAPF)**, a lightweight meta-learning framework that fuses frozen MARL expert policies through a Graph Convolutional Network (GCN) encoder and a Contextual Algorithm Selection (CAS) controller. GAPF encodes the sensor topology into a compact graph embedding at episode onset and selects the best-performing expert with only 1,473 trainable parameters. We evaluate GAPF across 7 scenarios spanning 20–100 sensors, 3 seeds each, totalling 21,000 test episodes. GAPF achieves near-perfect packet delivery ($\geq$98%) across all scales, with +43% to +111% relative improvement over standalone MARL baselines. It consumes 2–3.5$\times$ less energy, maintains 3–9$\times$ better energy balance, and requires 13$\times$ less memory (951 MB vs 13.2 GB), passing all feasibility constraints. The scalability analysis reveals that GAPF's advantage grows superlinearly in sparse networks, where topology-aware relay coordination is most critical. Our results demonstrate that lightweight policy fusion with graph-structural awareness can replace heavyweight monolithic MARL for resource-constrained, safety-critical WSN routing.

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Index Terms

wireless sensor networks, multi-agent reinforcement learning, graph neural networks, policy fusion, energy-efficient routing, disaster monitoring, QMIX, QTRAN, scalability, green AI.

I. Introduction

According to the European Environment Agency (EEA), a disaster is ”a serious disruption of the functioning of society, causing widespread human, material or environmental losses, which exceed the ability of affected society to cope using only its own resources” [1]. This scourge is a real handicap for our society according to statistics. Indeed, 398 natural disasters were recorded in 2023 with 95,000 deaths and losses of around 380 billion US dollars [2] [3]. Establishing a disaster monitoring policy thus appears to be essential.

Wireless Sensor Networks (WSNs) are emerging as an indispensable resource for disaster monitoring [4]. They owe this to their ability to collect data more accurately with minimal human intervention [5]. They are also resilient and maintain their connected state even if disasters have led to the destruction of infrastructure. In addition, they benefit from ease of scaling and configuration as required. In the case of wood fires, for example, detection methods such as watchtowers, satellite image processing methods, optical sensors, and methods based on digital cameras have disadvantages including inefficiency, consumption of energy, delay, accuracy, and implementation costs [6]. This is where WSNs prove valuable, being self-configuring, independent of infrastructure, efficient, and low in energy consumption. Despite the undeniable advantages of WSNs, large-scale issues need to be considered, notably battery life due to the problem of energy consumption [7]. It is therefore important to reduce the waste of energy. This becomes even more critical as rising energy consumption leads to higher CO2 emissions, which is harmful to the environment [8].

Many algorithms have been proposed to solve the energy efficiency problem in this context. In sum, we are able to highlight some limitations in the reviewed literature:

- A less realistic setting: In a disaster scenario, the wireless sensors static mobility consideration ([9], [10]) can fail since sensor position changes can be observed. Also, in a disaster setting, actions should be taken quickly to save lives and mitigate financial loss. For these reasons, real-time monitoring or at least near-real time monitoring should not be optional ([11], [12]).

- The reward function modeling: Some factors significantly influence network energy efficiency and can't be ignored. [13] propose a routing algorithm based on Q-learning to decrease and balance the energy consumed by sensors. The reward function in this work is a sum of partial rewards. This can be misleading since the relationship between the partial rewards and the final one could be complex and so a simple weighted sum cannot capture all this complexity.
- The theoretical limitations: In its current formulation, QMIX is limited from a theoretical point of view ([14], [15]). It is important to find the tradeoff between its good properties and some theoretical upgrade.

This work intends to overcome these limitations.

1) We propose **GAPF** (Graph-Attentive Policy Fusion), a lightweight meta-controller that sits on top of two independently pre-trained and frozen MADRL experts—QMIX and QTRAN—to capture the trade-off between a state-of-the-art value decomposition method and a theoretically more principled factorisation. GAPF encodes the WSN topology through a two-layer Graph Convolutional Network (GCN) and operates in two modes: *CAS* (Contextual Algorithm Selection), which selects the best-suited expert for an entire episode based on the graph embedding, and *Fusion*, which mixes the Q-value distributions of both experts at every time step via a learned softmax weighting. The meta-controller is trained end-to-end with REINFORCE using only $1{,}473$ parameters, making it extremely sample-efficient and deployable on resource-constrained devices.
2) We develop a bi-level reward structure built on a **Monotonic Attention Network**. At the inner level, each sensor receives a four-dimensional reward vector capturing (i) angular progress toward the base station, (ii) transmission energy consumption, (iii) dispersion of remaining energy across the network, and (iv) buffer occupancy. A query–key attention mechanism learns objective-specific weights $\alpha_i = \mathrm{softmax}(\mathbf{q} \cdot \mathbf{k}_i/\sqrt{d})$, and a monotonic

multilayer network ($4 \rightarrow 16 \rightarrow 16 \rightarrow 1$, all weights ≥ 0) scalarises the weighted vector into a single reward, guaranteeing that component-wise improvement always increases the scalar signal. At the outer level, the attention parameters are updated after each episode via **Evolution Strategies** to maximise a meta-objective J that combines energy efficiency and energy balance, decoupling reward shaping from the RL policy gradient.

3) We design a custom gym environment that is more realistic for a disaster scenario because it considers the dynamic mobility of nodes and factors which imbalance and increase the overall energy consumption of the network.

The rest of this paper is organized as follows. Section II gives the background and related work. Section III introduces the methodology of this work. Section IV analyzes and discusses the results. Section V presents the conclusion and describes the future work.

II. Background and Related Work

In recent years, a lot of research has been done on the routing problem. Moussa et al. [16] propose a Reinforcement Learning (RL) based routing protocol for Software Defined Network (SDN)-enabled WSN forest fire detection. The first step of the protocol was to design a clustering algorithm that introduces a delay in the re-clustering process to reduce energy. Secondly, an optimal path is defined by the SDN controller with the help of RL. Their work shows good results in network performance, however, it suffers from some drawbacks. Mainly, there is a lack of generalization in the RL model and a lack of being able to capture an important part in the factors which are involved in the reward construction.

Mishra et al. [17] propose the Energy-Balanced Grey Wolf Optimizer (EBGWO) for the clustering routing issue in Software-Defined WSN (SD-WSN). In their work, the selection of control nodes is tackled with the Gray-Wolf Optimization approach for energy balance. EBGWO succeeded in extending the network lifespan. The limitation however is the static consideration of the network [18].

Biabani et al. [19] provide a clustering and routing approach able to manage the nodes' energy consumption knowing the disaster environment characteristics. Their model is presented with hybrid Particle Swarm Optimization (PSO) and Harmony Search Algorithm to select the cluster head. Their multi-hop routing method is also based on PSO. Therefore, in disasters settings, the model is more efficient than the state-of-the-art approaches considering residual energy and other parameters. However, there are important delays regarding routing decisions [20] and the nodes are static in the network [19] which can be an issue in a disaster context.

Hosseinzadeh et al. [21] introduce a swarm intelligence-based clustering method to have cluster heads distributed more evenly. To select the cluster head and intermediate nodes for the routing process, authors use a firefly optimization approach and an aquila optimizer algorithm. This method improves the energy of the system. Nevertheless, the dynamic relocation of nodes needs to be explored [22].

Dehkordi et al. [23] propose a cluster-based routing model in WSN by using mobile sinks to decrease the nodes energy consumption and improve the network longevity. When comparing their method with other models, it comes at its superiority in terms of energy consumption among others. But the fact that mobile sinks are involved in their scheme, this may lead to some delays [22].

Gopalan et al. [24] elaborate a routing protocol where the Bacterial Foraging Optimization with Harmony Search Algorithm (BFO-HSA) is used for sensor nodes classification. The Cross-Layer-Based Opportunistic Routing Protocol (CORP) is considered to find the quickest path. Although the results demonstrate that their model outperforms the state of the art on certain quality of service parameters, it must also be recognized, as the authors point out, that the focus was on static sensor nodes [24].

Kodati et al. [25] propose a Hybrid Grasshopper and Harris Hawk Optimization Algorithm-based energy efficient routing protocol for optimal selection of clusters heads. At this effect, they consider a fitness function which considers many factors. Outcomes of this approach are better in terms of residual energy and throughput compared to the baseline methods. Yet, with this approach, there may be an energy imbalance [25].

Wang et al. [26] build an elite hybrid metaheuristic optimization algorithm-based routing algorithm to maximize the network lifetime of the WSN with a sink node. They successfully improve the survival time of the WSN in comparison to the state-of-the-art. However, the process of Cluster Head selection needs room for improvement [25].

Chaurasia et al. develop [27] a Meta-Heuristic Optimized Cluster Head Selection-Based Routing Algorithm for WSNs to minimize the energy consumption of sensors and increase the speed of the transmission of data. Underneath their model is a Dragonfly Algorithm for CH and path selection. It outperforms other models regarding the energy efficiency. However, the delay is higher [28].

Yao et al. [29] design an Energy-Efficient Routing Protocol based on Multi-Threshold Segmentation to improve the clustering process (cluster creation and the selection of CH). However, the CH selection is not properly done due to the problem of the overhead and clustering [30].

Manoharan et al. [31] propose an energy efficient routing in WSN using adaptive entropy bald eagle search optimization and density based adaptive soft clustering to prove the optimal data transfer in the network. The author's approach reaches better performance than other methods. A drawback is the non-consideration of mobile nodes [31].

Rai et al. [32] suggest a fuzzy Super Cluster Head (SCH) selection approach by considering four factors to reduce the dissipation of energy and improve the network longevity. Although it is true that this method outperforms some protocols, it should also be noted that under this approach there is a hot spot issue [33].

Prakash et al. [34] introduce an energy-efficient, cluster-based routing protocol — denoted SHO-CH — which leverages a Spotted Hyena Optimization strategy for cluster-head selection, and is tailored for heterogeneous wireless sensor networks (WSNs). SHO-CH optimize cluster formation and routing in order to reduce energy consumption across sensor nodes, thereby extending the operational lifetime of the network and enhancing energy-efficiency. The evaluation of the proposed model demonstrates improvements in stability period, overall network lifetime, and throughput compared with baseline protocols. However, the assumption of static sensor nodes and reliance on single-hop intra-cluster communication — as well as the computational overhead introduced by the iterative optimization process — limit the scheme's realism and applicability, especially in dynamic and unpredictable environments such as disaster-scenarios.

Ataoua et al. [35] introduce a novel clustering protocol for wireless sensor networks (WSNs), leveraging the Chernobyl Disaster Optimizer (CDO) metaheuristic to enhance energy efficiency and extend network lifetime. Their approach focuses on optimizing cluster-head selection and communication paths by minimizing intra-cluster distances and balancing energy consumption across nodes. Evaluated through simulations against standard protocols such as LEACH, the proposed method exhibits marked improvements in residual energy retention, reduced node failure rates, and prolonged network operation. Nonetheless, the study is constrained by its reliance on static simulation environments, omitting evaluation under disaster-specific dynamics such as node mobility, cascading failures, and communication interruptions. Additionally, the protocol overlooks key performance metrics including latency, packet delivery ratio, and security—factors essential to robust deployment in disaster monitoring contexts.

Qamar et al. [36] propose a hybrid routing framework that combines deterministic clustering with neural network–based energy prediction and energy-aware path selection to improve the efficiency and operational lifetime of wireless sensor networks (WSNs). The methodology integrates a static clustering mechanism derived from LEACH-D with an artificial neural network trained to estimate node energy levels, enabling routing decisions that prevent excessive load on low-energy nodes and promote balanced energy consumption across the network. Simulation results show reductions in energy consumption and notable improvements in network lifetime and packet delivery ratio compared with conventional protocols such as LEACH and related approaches. However, the framework assumes static network topologies and does not account for node mobility or failure dynamics commonly encountered in disaster scenarios. Furthermore, the neural network model is trained offline, which limits its adaptability to real-time environmental variations, and the absence of integrated security or trust mechanisms restricts its applicability in adversarial or disaster-prone deployments.

Rahman et al. [37] propose a blockchain-enabled architecture for wireless sensor networks (WSNs), aimed at detecting malicious nodes and preserving data integrity in critical scenarios such as disaster monitoring. The framework integrates a hybrid consensus model that combines smart contracts with a Proof of Information (PoI) mechanism to validate sensor data and isolate compromised nodes. Simulations report a data validity rate of 97.37%, highlighting the benefits of separating monitoring and control sensors to improve real-time responsiveness. However, the design does not incorporate energy-aware routing or clustering strategies, and its scalability in large-scale deployments remains unaddressed. Moreover, the protocol lacks evaluation under high-mobility or failure-prone conditions typical of disaster environments, limiting its practical applicability.

Rao et al. [30] propose a secure, cluster-based routing protocol for IoT-enabled wireless sensor networks (WSNs), incorporating a lightweight blockchain framework to ensure data integrity and resist tampering in decentralized deployments. The protocol integrates energy-aware clustering with hash-based data validation and smart contract mechanisms, aiming to enhance trustworthiness without compromising efficiency. Experimental evaluations report improved energy consumption, lower packet loss, and heightened data security relative to conventional routing approaches. However, the applicability of the proposed system to disaster monitoring remains limited, as the study is confined to agricultural use cases and does not address critical challenges such as node mobility, failure resilience, or intermittent connectivity. Furthermore, the added communication overhead introduced by the blockchain layer may hinder latency and scalability, especially in time-sensitive emergency response scenarios.

Sefati et al. [38] propose AFRL-HGS, a federated reinforcement learning framework combined with hybrid optimization techniques to support secure and adaptive routing in wireless sensor networks (WSNs). In this approach, sensor nodes act as reinforcement learning agents that collaboratively train routing policies through federated aggregation without sharing raw data, thereby preserving privacy and reducing communication overhead, while the hybrid optimization component accelerates convergence and stabilizes routing decisions. Simulation-based evaluation demonstrates improvements in packet delivery ratio, energy efficiency, and resilience to node failures compared with centralized reinforcement learning and conventional routing protocols. However, the framework assumes periodic synchronization and relatively stable connectivity for aggregating local models—conditions that may not hold in disaster environments characterized by network partitioning and high mobility—and it does not integrate additional trust or blockchain-based safeguards, potentially leaving the collaborative learning process vulnerable to model poisoning or adversarial manipulation.

Table I shows a comparative Analysis of Selected Routing Algorithms (2024–2025).

TABLE I: Comparative Analysis of Selected Routing Algorithms (2024–2025)

Article (Author, Year)	**Approach**	**Energy Efficiency**	**PDR**	**Latency**	**Mobility Support**
Ataoua et al. (2025)	Chernobyl Disaster Optimizer (CDO)-based clustering	High	Not evaluated	Not analyzed	No
Rahman et al. (2025)	Blockchain-secure routing with PoI consensus	Not addressed	High (97.37% data validity)	Moderate (blockchain overhead not quantified)	No
Qamar & Munir (2025)	Neural-assisted clustering with energy-aware routing	High	Improved	Not quantified	No
Rao et al. (2024)	Secure cluster-based routing with blockchain integrity checking	Moderate	Improved	Slightly increased due to blockchain	Limited
Sefati et al. (2025)	Federated RL with hybrid optimization (AFRL-HGS)	High	High	Low (fast convergence)	Partial (assumes stable sync)

This literature review shows the extensive work that has been done to achieve the energy efficiency goal of routing protocols and the limitations that they have faced which motivates our current work.

III. Methodology

A. Problem description

The basis of the proposed hybrid model consists of two Centralized Training and Decentralized Execution (CTDE) models named QMIX and QTRAN. As such, a formalization is needed in a context of MADRL. We assume there is a Base Station (BS) and N sensors nodes with the following characteristics:

$$S_k \equiv \begin{cases} CoverageRadius_k \ \ (Rad_k) \\ Position_k \\ RemainingEnergy_k \ \ (RE_k) \\ ConsumptionEnergy_k \ \ (CE_k) \\ Numberofpackets \ \ (N_k) \end{cases} \tag{1}$$

with Rad_k, $Position_k$, RE_k, CE_k, N_k being respectively the coverage radius, the coordinates in the environment to sense, the remaining energy, the consumption energy and the number of packets held by the k^{th} sensor.

Each sensor sends data to the base station through a multi hop process.

We assume that all sensors have the same coverage radius denoted R and at the beginning of each episode of the training, they have one packet to transmit. We set E_{max} to be the initial energy for all the sensors.

The observation space is defined as follows:

- Remaining energy for all sensors.
- Consumption energy for all sensors.
- Positions in the environment for all sensors: A two-dimensional space is considered.
- Number of packets for all sensors.

The routing objective is threefold:

1) **Maximize packet delivery ratio** — deliver as many sensor readings as possible to the base station, the primary goal of any data-gathering WSN;
2) **Minimize total energy consumption** — reduce the cumulative transmission and reception energy across all sensors to extend network operational lifetime;
3) **Balance energy usage across sensors** — minimize the standard deviation of remaining energy, preventing premature death of overloaded relay nodes.

These objectives are inherently conflicting: delivering more packets requires more transmissions, which increases energy consumption. The proposed framework addresses this tension through a bi-level optimization architecture. The *inner loop* (MARL policy) receives a scalar reward that combines all three objectives via four individual reward components — relay angle, energy consumption, energy dispersion, and receiver load — weighted by a Monotonic Q·K Attention Network. The *outer loop* (Evolution Strategies) tunes the attention network's parameters to maximize a meta-objective J that focuses on energy efficiency and energy balance, since packet delivery is already strongly incentivized by a dominant delivery bonus ($+100$) in the inner-loop reward. This separation ensures that the RL agents prioritize delivery during training while the scalarizer progressively learns to route energy-efficiently.

B. The Energy model

From [13] and [39], the energy model is defined as:

$$\begin{cases} E_T\left(M,d\right) = M \times a \times \left(E_{elec} + E_{amp} \times d^2\right) \\ \quad E_R\left(M\right) = M \times a \times E_{elec} \end{cases} \tag{2}$$

where $E_{elec} = 50\frac{nJ}{bit}$, $E_{amp} = 100\frac{pJ}{bit \times m^2}$, $E_T\left(M,d\right)$ is the energy that a sensor consumes to transmit M packets (each packet contains a amount of information) to a receiver at the distance d, $E_R\left(M\right)$ is the energy that a sensor consumes to receive M packets from the sender.

C. Reward Scalarization via Monotonic Q·K Attention Network

In cooperative MARL for WSN routing, agents must simultaneously optimize energy efficiency, energy balance, directional progress toward the base station, and load distribution. Fixed linear scalarizations require manual weight tuning that does not generalize across topologies. We introduce a *Monotonic Q·K Attention Network* that **(i)** learns objective importance weights via scaled dot-product attention [40], **(ii)** guarantees monotonicity so that Pareto-improving actions are never penalized [41], and **(iii)** adapts online through an Evolution Strategies (ES) outer loop [42].

1) Individual Reward Objectives: At each step t, when sensor i relays packets to neighbor j, the environment computes a four-dimensional reward vector $\mathbf{r}_i^{(t)} = [r_1, r_2, r_3, r_4] \in [0,1]^4$:

- **Directional progress** (r_1): $r_1 = 1 - |\theta_{ij}|/\pi$, where θ_{ij} is the angle between the relay direction $\overrightarrow{ij}$ and the ideal direction toward the base station (BS) $\overrightarrow{i\,\mathrm{BS}}$. Perfect alignment yields $r_1 = 1$; opposite direction yields $r_1 = 0$.
- **Energy efficiency** (r_2): $r_2 = 1-(E_{\mathrm{tx}}(p_i, d_{ij})+E_{\mathrm{rx}}(p_i))/(E_{\mathrm{tx}}^{\max}+E_{\mathrm{rx}}^{\max})$, where $E_{\mathrm{tx}}(p_i, d_{ij})$ and $E_{\mathrm{rx}}(p_i)$ are the transmission and reception energies for p_i packets over distance d_{ij}, normalized by worst-case values $E_{\mathrm{tx}}^{\max} = E_{\mathrm{tx}}(N \cdot p_0, R)$ and $E_{\mathrm{rx}}^{\max} = E_{\mathrm{rx}}(N \cdot p_0)$, with N the number of sensors, p_0 the initial packet count per sensor, and R the communication radius.
- **Energy balance** (r_3): $r_3 = 1-\sigma(\mathbf{e}_{\mathrm{rem}})/(E_0/2)$, where $\sigma(\mathbf{e}_{\mathrm{rem}})$ is the standard deviation of remaining energies across all sensors, E_0 is the initial energy budget per sensor, and $E_0/2$ is the theoretical maximum standard deviation (when half the sensors are fully charged and half depleted).
- **Load balancing** (r_4): $r_4 = 1-p_j/(N \cdot p_0)$, where p_j is the current packet count at receiver node j.

All four components are clipped to $[0, 1]$.

2) Monotonic Q·K Attention Architecture: The scalarization function $f_\phi : [0, 1]^4 \to \mathbb{R}$ maps the reward vector $\mathbf{r}$ to a scalar via two stages.

a) Stage 1: Attention weighting.: A learnable query vector $\mathbf{q} \in \mathbb{R}^d$ and a key matrix $\mathbf{K} = [\mathbf{k}_1, \ldots, \mathbf{k}_4]^\top \in \mathbb{R}^{4\times d}$ (with embedding dimension $d = 16$) produce attention weights $\alpha_i = \mathrm{softmax}(\mathbf{K}\mathbf{q}/\sqrt{d})_i$, yielding the weighted input $\mathbf{r}' = \boldsymbol{\alpha}\odot\mathbf{r}$ (element-wise product). Because $\boldsymbol{\alpha}$ depends only on the learnable parameters $(\mathbf{q}, \mathbf{K})$ and not on $\mathbf{r}$, and since $\alpha_i > 0$ for all i (due to softmax), the weighting preserves the component-wise partial order: $\mathbf{r}^{(1)} \leq \mathbf{r}^{(2)}$ implies $\mathbf{r}'^{(1)} \leq \mathbf{r}'^{(2)}$.

b) Stage 2: Monotonic network.: The weighted input $\mathbf{r}'$ passes through a three-layer feed-forward network ($4 \to 16 \to 16 \to 1$) with softplus activations $\sigma(x) = \log(1 + e^x)$ and non-negative weight matrices enforced via $\mathbf{W}_\ell = \mathrm{softplus}(\mathbf{W}_\ell^{\mathrm{raw}}) \geq 0$. Since the softplus derivative $\sigma'(x) = (1 + e^{-x})^{-1} \in (0, 1)$ is strictly positive and all weights are non-negative, the Jacobian satisfies $\partial f/\partial r_i \geq 0$ for every component. By the chain rule over all layers, the full network

$f_\phi = g_\psi \circ (\boldsymbol{\alpha} \odot \cdot)$ is monotonically non-decreasing: any Pareto-dominating action always receives a higher scalar reward. The output is centered as $\hat{f}_\phi(\mathbf{r}) = f_\phi(\mathbf{r}) - f_\phi(\mathbf{0})$ to improve signal-to-noise ratio, where the baseline $f_\phi(\mathbf{0})$ is recomputed once per episode.

3) Composite Step Reward: The final reward for sensor i at step t depends on its action: **(a)** delivery to the BS yields $r_i^{(t)} = \beta_{\text{del}} \cdot p_i$, where $\beta_{\text{del}} = 100$ is a delivery bonus and p_i the number of packets delivered (bypassing scalarization, as delivery is the terminal objective); **(b)** relay to neighbor $j \neq \text{BS}$ yields:

$$r_i^{(t)} = \underbrace{\hat{f}_\phi(\mathbf{r}_i^{(t)}) \cdot \lambda_{\text{attn}}}_{\text{attention-scalarized}} + \underbrace{\frac{d(i, \text{BS}) - d(j, \text{BS})}{R} \cdot 0.5}_{\text{progress shaping}} \tag{3}$$

where $\lambda_{\text{attn}} = 0.01$ scales the attention output to prevent relay-loop farming, and $d(\cdot, \text{BS})$ denotes the Euclidean distance to the BS; **(c)** invalid actions receive penalties of -0.1 (self-loops, out-of-range) to -0.5 (insufficient energy); inactive sensors receive 0.

4) Evolution Strategies Meta-Learning: The attention network parameters $\phi = (\mathbf{q}, \mathbf{K}, \{\mathbf{W}_\ell^{\text{raw}}, \mathbf{b}_\ell\}_\ell)$ are trained via an outer-loop ES [42] operating at episode granularity, without differentiating through the RL episode. At episode end, a meta-objective $J \in [0, 2]$ evaluates the true network-level goals:

$$J = \underbrace{\left(1 - \frac{\sum_{i=1}^{N}(E_0 - e_i^{\text{rem}})}{N \cdot E_0}\right)}_{\text{energy efficiency}} + \underbrace{\left(1 - \frac{\sigma(\mathbf{e}_{\text{rem}})}{E_0/2}\right)}_{\text{energy balance}} \tag{4}$$

where e_i^{rem} is the remaining energy of sensor i at episode end. Each episode: **(1)** parameters are perturbed as $\phi' = \phi + \sigma_{\text{ES}}\,\boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and noise scale $\sigma_{\text{ES}} = 0.01$; **(2)** the episode executes with ϕ' and $J(\phi')$ is computed; **(3)** the advantage $A = J(\phi') - \bar{J}$ is evaluated against an exponential moving average baseline $\bar{J}$ (decay $\beta = 0.9$), and parameters are updated as $\phi \leftarrow \phi + \eta_{\text{ES}} \cdot A \cdot \boldsymbol{\varepsilon}$ with learning rate $\eta_{\text{ES}} = 0.01$ (skipped when $|A| < 0.01$). This bi-level optimization decouples inner-loop RL training from outer-loop reward shaping. Key embeddings for energy ($\mathbf{k}_2$) and dispersion ($\mathbf{k}_3$) are initialized close to $\mathbf{q}$ (higher initial attention), encoding the prior that energy-related objectives are most important, while the ES procedure discovers the optimal weighting. The learned weights $\boldsymbol{\alpha}$ provide direct interpretability of objective importance at any point during training.

D. GAPF: Graph-Attentive Policy Fusion

We introduce **GAPF** (Graph-Attentive Policy Fusion), a lightweight meta-algorithm that selects, at the episode level, the best-performing MARL expert for a given WSN topology. Inspired by adaptive-policy approaches that train multiple policies and defer selection to execution time [43], and by recent work demonstrating the effectiveness of GNNs for wireless network optimisation [44], GAPF uses a graph neural network (GNN) to encode the spatial structure of the sensor network and a meta-controller trained via REINFORCE [45] to choose between two pre-trained, frozen experts (QMIX and QTRAN).

1) Problem Formulation: Let $\mathcal{E} = \{\pi_{\text{QMIX}}, \pi_{\text{QTRAN}}\}$ denote pre-trained, frozen MARL policies for the WSN routing Dec-POMDP. At the start of each episode k, the WSN defines a communication graph $\mathcal{G}_k = (\mathcal{V}, \mathcal{E}_k)$, where nodes $\mathcal{V} = \{1, \ldots, N\}$ are sensors and an edge $(i, j) \in \mathcal{E}_k$ exists iff $\|p_i - p_j\|_2 \leq R$, with $p_i \in \mathbb{R}^2$ the position of sensor i and R the communication radius. GAPF learns a parameterised policy μ_θ mapping the topology $\mathcal{G}_k$ to a selection $a_k^{\text{meta}} \in \{\text{QMIX}, \text{QTRAN}\}$, maximising the expected episodic return:

$$\max_\theta \ \mathbb{E}_{\mathcal{G}_k}\left[G_k \,\middle|\, \pi_{a_k^{\text{meta}}} \right], \quad a_k^{\text{meta}} \sim \mu_\theta(\cdot \mid \mathcal{G}_k). \tag{5}$$

2) Architecture: The GAPF model μ_θ consists of a *GCN encoder* producing a fixed-dimensional graph embedding, and a *CAS controller* (Contextual Algorithm Selection) mapping this embedding to a Bernoulli selection probability (Figure 1).

a) Graph construction.: Given the N sensor positions $\{p_i\}_{i=1}^N$ and communication radius R, the binary adjacency matrix $\mathbf{A} \in \{0, 1\}^{N \times N}$ is defined as $A_{ij} = \mathbf{1}[\|p_i - p_j\| \leq R] \cdot \mathbf{1}[i \neq j]$, then symmetrically normalised following [46]:

$$\hat{\mathbf{A}} = \tilde{\mathbf{D}}^{-1/2}\, \tilde{\mathbf{A}}\, \tilde{\mathbf{D}}^{-1/2}, \quad \tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}_N, \quad \tilde{D}_{ii} = \textstyle\sum_j \tilde{A}_{ij}. \tag{6}$$

Each sensor i is described by a feature vector $\mathbf{x}_i \in \mathbb{R}^4$:

$$\mathbf{x}_i = \left[\frac{p_i^{(x)}}{L}, \ \frac{p_i^{(y)}}{L}, \ \frac{\|p_i - p_{\mathrm{BS}}\|}{\max_j \|p_j - p_{\mathrm{BS}}\|}, \ \frac{\deg(i)}{\max_j \deg(j)} \right]^\top, \tag{7}$$

where L is the field side length, p_{BS} is the base station position, and $\deg(i) = \sum_j A_{ij}$ is the node degree. All features are normalised to $[0, 1]$.

The complete graph construction procedure is formalised in Algorithm 1.

Algorithm 1 Graph Construction — Pseudocode

Require: Sensor positions $\{p_i\}_{i=1}^N$, base station position p_{BS}, communication radius R, field side length L

Ensure: Normalised adjacency $\hat{\mathbf{A}} \in \mathbb{R}^{N \times N}$, node feature matrix $\mathbf{X} \in \mathbb{R}^{N \times 4}$

Initialise $\mathbf{A} \leftarrow \mathbf{0}_{N \times N}$ {Binary adjacency matrix}
for each pair (i, j) with $i \neq j$ and $\|p_i - p_j\| \leq R$ **do**
    $A_{ij} \leftarrow 1$
end for
$\tilde{\mathbf{A}} \leftarrow \mathbf{A} + \mathbf{I}_N$; compute $\tilde{D}_{ii} \leftarrow \sum_j \tilde{A}_{ij}$ for all i {Self-loops + degree}
$\hat{\mathbf{A}} \leftarrow \tilde{\mathbf{D}}^{-1/2} \, \tilde{\mathbf{A}} \, \tilde{\mathbf{D}}^{-1/2}$ {Symmetric normalisation}
$d_{\max} \leftarrow \max_j \|p_j - p_{\mathrm{BS}}\|$; $\quad \deg_{\max} \leftarrow \max_j \deg(j)$
for each sensor $i = 1, \ldots, N$ **do**
    $\mathbf{x}_i \leftarrow [p_i^{(x)}/L, \ p_i^{(y)}/L, \ \|p_i - p_{\mathrm{BS}}\|/d_{\max}, \ \deg(i)/\deg_{\max}]^\top$ {Normalised features}
end for
$\mathbf{X} \leftarrow [\mathbf{x}_1, \ldots, \mathbf{x}_N]^\top$
return $\hat{\mathbf{A}}, \ \mathbf{X}$

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b) GCN encoder.: The node feature matrix $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_N]^\top \in \mathbb{R}^{N\times 4}$ is processed by a two-layer GCN [46]:

$$\mathbf{H}^{(1)} = \mathrm{ReLU}(\hat{\mathbf{A}}\,\mathbf{X}\,\mathbf{W}_1), \quad \mathbf{Z} = \hat{\mathbf{A}}\,\mathbf{H}^{(1)}\,\mathbf{W}_2, \quad \mathbf{z} = \tfrac{1}{N}\textstyle\sum_{i=1}^{N} \mathbf{Z}_{i,:} \in \mathbb{R}^d \tag{8}$$

where $\mathbf{W}_1 \in \mathbb{R}^{4\times d}$, $\mathbf{W}_2 \in \mathbb{R}^{d\times d}$, and $d = 16$ is the embedding dimension. The mean-pool readout is permutation-invariant and handles variable numbers of sensors without architectural changes.

c) CAS controller.: The graph embedding $\mathbf{z}$ is fed to a two-layer MLP that outputs the probability of selecting QMIX:

$$p_{\text{QMIX}} = \sigma\big(\mathbf{w}_2^\top\,\mathrm{ReLU}(\mathbf{W}_c\,\mathbf{z} + \mathbf{b}_c) + b_2\big)\,, \tag{9}$$

where $\mathbf{W}_c \in \mathbb{R}^{h\times d}$, $\mathbf{b}_c \in \mathbb{R}^h$, $\mathbf{w}_2 \in \mathbb{R}^h$, $b_2 \in \mathbb{R}$, $h = 64$ is the hidden dimension, and σ is the sigmoid function. During training, $a^{\text{meta}} \sim \text{Bernoulli}(p_{\text{QMIX}})$; at evaluation, $a^{\text{meta}} = \text{QMIX}$ if $p_{\text{QMIX}} > 0.5$, QTRAN otherwise. The total number of trainable parameters is:

$$|\theta| = \underbrace{4\times 16 + 16\times 16}_{\text{GCN: 320}} + \underbrace{16\times 64 + 64 + 64\times 1 + 1}_{\text{CAS: 1,153}} = 1{,}473. \tag{10}$$

This is approximately $26\times$ smaller than a single RNN agent (38,983 parameters), making GAPF negligible in computational overhead.

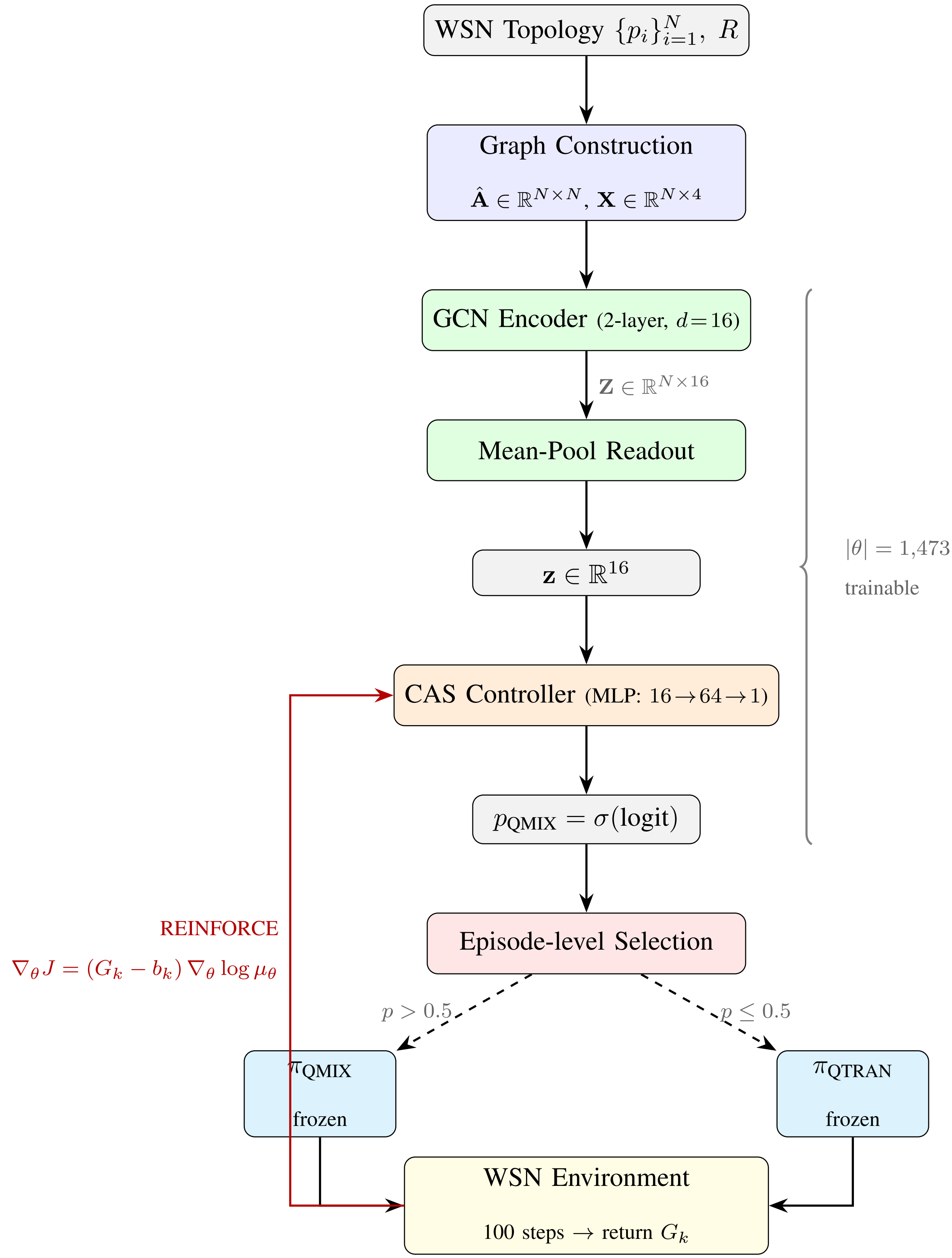
WSN Topology $\{p_i\}_{i=1}^{N}, \ R$
Graph Construction
$\hat{\mathbf{A}} \in \mathbb{R}^{N\times N}$, $\mathbf{X} \in \mathbb{R}^{N\times 4}$
GCN Encoder (2-layer, $d=16$)
$\mathbf{Z} \in \mathbb{R}^{N\times 16}$
Mean-Pool Readout
$\mathbf{z} \in \mathbb{R}^{16}$
$|\theta| = 1{,}473$
trainable
CAS Controller (MLP: $16\to 64\to 1$)
$p_{\text{QMIX}} = \sigma(\text{logit})$
REINFORCE
$\nabla_\theta J = (G_k - b_k)\,\nabla_\theta \log \mu_\theta$
Episode-level Selection
$p > 0.5$
$p \leq 0.5$
π_{QMIX}
frozen
π_{QTRAN}
frozen
WSN Environment
100 steps $\to$ return G_k

Fig. 1: GAPF architecture (CAS mode). Sensor positions and communication radius define a graph encoded by a two-layer GCN with mean-pool readout ($\mathbf{z} \in \mathbb{R}^{16}$). The CAS controller maps $\mathbf{z}$ to a selection probability p_{QMIX}. The chosen frozen expert executes the full episode. The return G_k feeds back (red arrow) as a REINFORCE signal to update only the $1{,}473$ meta-controller parameters.

3) Training Procedure: QMIX and QTRAN are trained independently; their parameters are then frozen. Only the GAPF meta-controller ($1{,}473$ parameters) is trained via REINFORCE [45] with an EMA baseline. Let G_k denote the episodic return from executing the selected expert $\pi_{a_k^{\text{meta}}}$ in episode k. The policy gradient is:

$$\nabla_\theta J(\theta) = \mathbb{E}[(G_k - b_k)\, \nabla_\theta \log \mu_\theta(a_k^{\text{meta}} \mid \mathcal{G}_k)]\,, \tag{11}$$

where b_k is the EMA baseline updated as $b_{k+1} = 0.95\, b_k + 0.05\, G_k$. Since the selection is Bernoulli, $\log \mu_\theta = \log p_{\text{QMIX}}$ if QMIX is chosen and $\log(1 - p_{\text{QMIX}})$ otherwise. Parameters are updated via Adam with learning rate $\alpha = 3 \times 10^{-3}$ and gradient clipping $\|\nabla_\theta\|_2 \leq 1.0$. The complete procedure is summarised in Algorithm 2.

Algorithm 2 GAPF Training — Pseudocode (CAS Mode)

Require: Frozen expert policies π_{QMIX}, π_{QTRAN}; WSN environment $\mathcal{M}$; training episodes K

Ensure: Trained meta-controller parameters θ

{— Initialisation —}
Initialise θ using Xavier uniform distribution
Create Adam optimiser with learning rate $\alpha = 3 \times 10^{-3}$
$b \leftarrow$ None {EMA baseline for variance reduction}
for $k = 1, \ldots, K$ **do**
    {— Topology encoding (once per episode) —}
    Reset $\mathcal{M}$; obtain sensor positions $\{p_i\}$ and radius R
    $(\hat{\mathbf{A}}_k, \mathbf{X}_k) \leftarrow$ Algorithm 1 applied to $(\{p_i\}, p_{\text{BS}}, R, L)$
    $\mathbf{H} \leftarrow \text{ReLU}(\hat{\mathbf{A}}_k \cdot \mathbf{X}_k \cdot \mathbf{W}_1)$ {GCN layer 1}
    $\mathbf{Z} \leftarrow \hat{\mathbf{A}}_k \cdot \mathbf{H} \cdot \mathbf{W}_2$ {GCN layer 2}

$\mathbf{z}_k \leftarrow \frac{1}{N}\sum_{i=1}^{N}\mathbf{Z}_{i,:}$ {Mean-pool graph embedding}
{— Expert selection (stochastic during training) —}
$p_{\text{QMIX}} \leftarrow \sigma(\mathbf{w}_2^{\top} \cdot \text{ReLU}(\mathbf{W}_c \cdot \mathbf{z}_k + \mathbf{b}_c) + b_2)$ {CAS probability}
Sample $a_k^{\text{meta}} \sim \text{Bernoulli}(p_{\text{QMIX}})$ $\{1 \rightarrow$ QMIX, $0 \rightarrow$ QTRAN$\}$
Execute expert $\pi_{a_k^{\text{meta}}}$ for T steps; collect episodic return G_k
{— REINFORCE update with EMA baseline —}
if $b =$ None **then**
    $b \leftarrow G_k$ {Initialise baseline with first return}
end if
$\delta_k \leftarrow G_k - b$ {Advantage estimate}
$b \leftarrow 0.95 \cdot b + 0.05 \cdot G_k$ {Update EMA baseline}
{Skip update if advantage is near zero}
if $|\delta_k| > 10^{-8}$ **then**
    $\mathcal{L} \leftarrow -\delta_k \cdot \log \mu_\theta(a_k^{\text{meta}} \mid \mathcal{G}_k)$ {Policy gradient loss}
    Clip gradients: $\|\nabla_\theta \mathcal{L}\|_2 \leq 1.0$
    Update θ via Adam
end if
end for
return θ

Algorithm 3 describes the full inference-time procedure used during evaluation and deployment.

Algorithm 3 GAPF Episode Execution — Pseudocode (Inference)

Require: Trained meta-controller μ_θ; frozen experts π_{QMIX}, π_{QTRAN}; environment $\mathcal{M}$; episode length T

Ensure: Episodic return G, routing metrics (PDR, E_{tot}, σ_E)

{— Phase 1: Topology encoding (executed once per episode) —}
Reset $\mathcal{M}$; obtain sensor positions $\{p_i\}_{i=1}^{N}$ and radius R
$(\hat{\mathbf{A}}, \mathbf{X}) \leftarrow$ Algorithm 1 applied to $(\{p_i\}, p_{\text{BS}}, R, L)$
$\mathbf{H} \leftarrow \text{ReLU}(\hat{\mathbf{A}} \cdot \mathbf{X} \cdot \mathbf{W}_1)$ {GCN layer 1: neighbour aggregation}
$\mathbf{Z} \leftarrow \hat{\mathbf{A}} \cdot \mathbf{H} \cdot \mathbf{W}_2$ {GCN layer 2}
$\mathbf{z} \leftarrow \frac{1}{N} \sum_{i=1}^{N} \mathbf{Z}_{i,:}$ {Mean-pool readout $\rightarrow \mathbf{z} \in \mathbb{R}^{16}$}
{— Phase 2: Expert selection via CAS controller —}
$p_{\text{QMIX}} \leftarrow \sigma\big(\mathbf{w}_2^{\top} \cdot \text{ReLU}(\mathbf{W}_c \cdot \mathbf{z} + \mathbf{b}_c) + b_2\big)$ {Selection probability}
if $p_{\text{QMIX}} > 0.5$ **then**
    SelectedExpert $\leftarrow \pi_{\text{QMIX}}$
else
    SelectedExpert $\leftarrow \pi_{\text{QTRAN}}$
end if
{— Phase 3: Decentralised step-by-step execution —}
Initialise hidden states $\mathbf{h}_i^{(0)} \leftarrow \mathbf{0}$ for all agents i
$G \leftarrow 0$
for $t = 1, \ldots, T$ **do**
    for each agent $i = 1, \ldots, N$ **do**
        Observe local state $\mathbf{o}_i^{(t)} \leftarrow [RE_i,\ CE_i,\ x_i,\ y_i,\ N_i]$ {5-dim observation}
        $(\mathbf{q}_i,\ \mathbf{h}_i^{(t)}) \leftarrow$ SelectedExpert$(\mathbf{o}_i^{(t)},\ \mathbf{h}_i^{(t-1)})$ {GRU forward pass}
        Set $q_{ij} \leftarrow -\infty$ for out-of-range or energy-depleted neighbour j {Valid-action mask}
        $a_i^{(t)} \leftarrow \arg\max_j \mathbf{q}_i[j]$ {Greedy action selection}
    end for
    Execute joint action $(a_1^{(t)}, \ldots, a_N^{(t)})$ in $\mathcal{M}$
    Collect rewards $\{r_i^{(t)}\}$; update environment state
    $G \leftarrow G + \sum_{i=1}^{N} r_i^{(t)}$
end for
Compute final metrics: PDR, E_{tot}, σ_E from environment state

29: **return** G, PDR, E_{tot}, σ_E

---

4) Design Rationale: Five design choices distinguish GAPF from standard ensemble methods and recent mixture-of-experts RL architectures [47]: **(1)** *Topology awareness*: the GCN encodes the spatial graph, allowing the controller to learn that certain topologies favour one expert over another. **(2)** *Episode-level granularity*: selection occurs once per episode, avoiding instability from switching experts mid-episode where GRU-based agents have incompatible hidden state dynamics. **(3)** *Minimal overhead*: $1{,}473$ parameters vs. $38{,}983$ per expert; the GCN runs once per episode and the CAS decision is a single MLP forward pass. **(4)** *Frozen experts*: pre-trained policies are not fine-tuned, preserving validated performance and reducing the optimisation landscape to $1{,}473$ dimensions. **(5)** *Deployment feasibility*: each agent's GRU ($38{,}983$ parameters ≈ 38 KB in int8 quantisation) fits within typical WSN microcontroller RAM (nRF52840: 256 KB; ESP32: 520 KB), enabling on-device inference.

5) Comparison with Alternative Lightweight Approaches: Strategies based on node-degree or local graph metrics (clustering coefficient, betweenness centrality) operate on *pre-defined, single-hop* structural features that cannot encode higher-order topological patterns such as bottleneck corridors or multi-hop connectivity islands. In contrast, GAPF's two-layer GCN propagates information across each node's 2-hop neighbourhood, producing a graph-level embedding $\mathbf{z} \in \mathbb{R}^{16}$ that captures global connectivity structure; notably, node-degree is already one of the four input features (Eq. 7), which the GCN *enriches* through aggregation. Fuzzy inference systems (e.g., QPSOFL's Mamdani-type controller) require manual specification of linguistic variables, membership functions, and rule bases—a topology-specific process that does not transfer across configurations without re-tuning. GAPF avoids this feature engineering bottleneck: its GCN weights are learned end-to-end via REINFORCE and generalise across 7 scenarios (20–100 sensors, 25–50 m radius) at $\geq$98% PDR without per-scenario tuning, whereas QPSOFL collapses below 15% PDR in sparse topologies. The GCN's per-episode overhead ($\mathcal{O}(n^2 d)$, d=16, < 0.03 ms) is negligible relative to per-step routing costs.

6) Meta-Learning Architecture vs. End-to-End Multi-Agent DRL: GAPF's two-stage design (pre-trained frozen experts + lightweight meta-controller) contrasts with end-to-end multi-agent DRL along three dimensions. *Learning dynamics*: end-to-end MARL must simultaneously optimise agent policies, value decomposition, and implicit topology adaptation through a single gradient signal, creating a non-stationary problem where each agent's optimal policy depends on every other agent's changing policy; GAPF separates these concerns—experts converge independently, and the 1,473-parameter meta-controller solves the simpler problem of mapping topology embeddings to binary selection, converging in $\sim$2,000 episodes versus $\sim$200,000 steps per expert. *Memory*: the dominant cost in end-to-end MARL is the replay buffer (15,000 episodes $\times$ n agents); GAPF trains on-policy and eliminates this entirely, yielding a $13\times$ reduction in peak RAM (Table VIII) and passing the 1,024 MB constraint that standalone QMIX/QTRAN fail in every scenario. *Per-step computation*: at execution time, GAPF's per-step cost is identical to standalone QMIX or QTRAN ($\mathcal{O}(n^2h + nh^2)$); the GCN adds only a one-time $\mathcal{O}(n^2d)$ cost amortised over T=100 steps.

Here is the complete time-complexity analysis.

E. Time Complexity Analysis

Let n denote the number of sensor agents, $h = 64$ the RNN hidden dimension, $d = 16$ the GNN embedding dimension, and T the number of steps per episode.

At each routing step, only the frozen expert executes; across all n agents the per-step cost is $\mathcal{O}_{\text{step}} = \mathcal{O}(n^2h + nh^2)$, **identical for QMIX, QTRAN, and GAPF**. GAPF adds a one-time per-episode overhead for adjacency construction ($\mathcal{O}(n^2)$), a two-layer GCN forward pass ($\mathcal{O}(n^2d + nd^2)$), and the CAS MLP ($\mathcal{O}(dm)$), yielding $\mathcal{O}_{\text{GAPF-init}} = \mathcal{O}(n^2d)$. For an episode of T steps the total is $\mathcal{O}(Tn^2h + Tnh^2 + n^2d)$; since $Th \gg d$ the GCN overhead is amortised and the dominant term $\mathcal{O}(Tn^2h)$ matches standalone QMIX/QTRAN.

TABLE II: Time complexity comparison. n: number of sensors, T: steps per episode, h: RNN hidden dimension (64), d: GNN embedding dimension (16), N_p: PSO particles (32), I: PSO iterations (50).

Algorithm	Per-Step	Per-Episode
QMIX / QTRAN	$\mathcal{O}(n^2h + nh^2)$	$\mathcal{O}(T(n^2h + nh^2))$
QPSOFL	$\mathcal{O}(N_p \cdot n^2)$	$\mathcal{O}(T \cdot I \cdot N_p \cdot n^2)$
GAPF	$\mathcal{O}(n^2h + nh^2)$	$\mathcal{O}(T(n^2h + nh^2) + n^2d)$

GAPF adds only $\mathcal{O}(n^2d)$ per episode, where $d = 16 \ll h = 64$, making its per-step complexity identical to standalone QMIX/QTRAN. In contrast, QPSOFL's cost grows as $\mathcal{O}(T \cdot I \cdot N_p \cdot n^2)$ due to repeated swarm optimisation at every step, explaining its longer wall-clock times despite having no neural network.

IV. Results and Discussion

Table III presents the custom Gym environment setting.

TABLE III: Custom Gym Environment setting.

Key	**Value**
E_{elec}	50×10^{-9} J/bit
E_{amp}	100×10^{-12} J/(bit·m^2)
Packet size	512 bit
Initial energy per sensor (E_0)	1 J
Field bounds (LowerBound, UpperBound)	0–100 m
Base station position	(50, 50)
Initial packets per sensor	1
Total timesteps / Max per episode	200,000 / 100

Table IV displays the training hyperparameters. QMIX and QTRAN are trained independently as separate expert policies; their parameters are then frozen, and only the lightweight GAPF meta-controller ($\sim$1,473 parameters) is trained to dynamically select between them based on network topology.

TABLE IV: Training Hyperparameters for QMIX, QTRAN, and GAPF.

Category	**Parameter**	**QMIX**	**QTRAN**	**GAPF**
Optimization	Optimizer / LR	Adam / 0.0003	RMSprop / 0.0003	Adam / 0.003
	Grad clip / γ	10 / 0.99	10 / 0.99	1.0 / —
	Reward scale	0.01	0.01	—
Exploration	Strategy	ε-greedy	ε-greedy	REINFORCE
	ε: start→end (anneal)	$0.5 \to 0.01$ (150K)	$0.5 \to 0.01$ (150K)	—
Architecture	Agent / hidden dim	GRU(64) / 64	GRU(64) / 64	GCN+MLP / 16+64
	Mixing embed dim	64	64	—
Training	Episodes / batch	∼2K / 32	∼2K / 32	2K / 1 (on-policy)
	Replay buffer	15,000	15,000	—
	Target update	Hard, every 30 ep.	Hard, every 30 ep.	EMA β=0.95
QTRAN-specific	λ_{opt} / λ_{nopt}	—	1.0 / 0.1	—
Evaluation	Test episodes / eval ε	1,000 / 0.01	1,000 / 0.01	1,000 / —

Our evaluation includes QPSOFL [48], a metaheuristic baseline using Quantum PSO with Sobol-initialized particles and L'evy flight for cluster-head selection, followed by Mamdani-type Fuzzy Inference for relay selection.

a) Execution-time instrumentation.: The framework records per-step wall-clock times (mean, p95, p99) as versioned `.npy` arrays. A standalone benchmark tool measures latency over 30 iterations (3 warmup), peak RAM via `tracemalloc`, analytical MACs/FLOPs, and estimated energy per inference (μJ). Table V reports results for the worst case (N=70).

TABLE V: Inference metrics (Apple M-series CPU, 0.1,W reference). All algorithms achieve sub-millisecond latency.

Algorithm	**Latency p95 (ms)**	**Peak RAM (KB)**	**MACs**	**Energy (μJ)**
QMIX (agent + mixer)	0.27	2.98	2,539,904	27.2
QTRAN (agent + mixer)	0.17	34.32	4,955,532	16.7
GAPF (GCN, gateway)	0.03	4.12	181,408	3.0
Agent only (deployed)	<1	∼2	∼29K	<10

Even scaling by $50\times$ for a resource-constrained ARM Cortex-M4 at 80,MHz, the per-step decision time remains under 15,ms—well within real-time WSN budgets.

Seven scenarios (Table VI) systematically vary network density (n) and coverage radius (R), each evaluated over three random seeds (42, 123, 456).

TABLE VI: Overview of the seven evaluation scenarios.

#	n	R (m)	**Rationale**
1	20	25	Sparse, short range — extreme connectivity challenge
2	30	25	Small–medium, short range — limited relay options
3	50	35	Medium, standard range — moderate density
4	70	35	Baseline — dense, standard range
5	70	45	Dense, extended range — more routing options
6	100	35	Large-scale, standard range — scalability test
7	100	50	Large-scale, long range — high connectivity stress

1) Consolidated Results Across All Scenarios: Table VII presents key performance metrics for all seven scenarios. All values are mean $\pm$ std aggregated over three seeds on the 1,000-episode test set. PDR denotes packet delivery ratio, E_{tot} total energy consumed (J), σ_E standard deviation of remaining energy (lower is better balanced), and Throughput is packets delivered per step.

TABLE VII: Performance metrics across all scenarios (three seeds, test set). Best values per scenario in **bold**.

		PDR				**E_{tot} (J)**			
n	R	QMIX	QTRAN	QPSOFL	**GAPF**	QMIX	QTRAN	QPSOFL	**GAPF**
20	25	$.559 \pm .257$	$.558 \pm .258$	$.075 \pm .107$	$\mathbf{.753 \pm .161}$	$.093 \pm .055$	$.093 \pm .055$	$\mathbf{.052 \pm .049}$	$.075 \pm .044$
30	25	$.568 \pm .179$	$.567 \pm .174$	$.053 \pm .046$	$\mathbf{.877 \pm .159}$	$.165 \pm .054$	$.166 \pm .054$	$\mathbf{.102 \pm .050}$	$.102 \pm .070$
50	35	$.553 \pm .248$	$.552 \pm .250$	$.153 \pm .092$	$\mathbf{.990 \pm .088}$	$.423 \pm .152$	$.422 \pm .153$	$\mathbf{.113 \pm .072}$	$.202 \pm .131$
70	35	$.545 \pm .242$	$.545 \pm .243$	$.122 \pm .090$	$\mathbf{.980 \pm .119}$	$.600 \pm .204$	$.603 \pm .200$	$\mathbf{.198 \pm .116}$	$.283 \pm .213$
70	45	$.667 \pm .246$	$.659 \pm .256$	$.365 \pm .092$	$\mathbf{.980 \pm .124}$	$.687 \pm .265$	$.692 \pm .274$	$\mathbf{.040 \pm .040}$	$.334 \pm .272$
100	35	$.464 \pm .270$	$.462 \pm .261$	$.099 \pm .080$	$\mathbf{.980 \pm .124}$	$1.010 \pm .295$	$1.017 \pm .282$	$\mathbf{.312 \pm .171}$	$.334 \pm .272$
100	50	$.669 \pm .266$	$.684 \pm .270$	$.448 \pm .057$	$\mathbf{.980 \pm .124}$	$1.181 \pm .413$	$1.180 \pm .412$	$\mathbf{.026 \pm .025}$	$.334 \pm .272$

TABLE VIII: Energy balance (σ_E) and feasibility compliance across all scenarios.

		σ_E (lower = better)				Memory < 1024,MB?		$p_{99} < 50$,ms?	
n	R	QMIX	QTRAN	QPSOFL	**GAPF**	QMIX/QTRAN	GAPF	QMIX/QTRAN	GAPF
20	25	.0075	.0075	.0056	**.0043**	FAIL	PASS	PASS	PASS
30	25	.0111	.0112	.0085	**.0035**	FAIL	PASS	PASS	PASS
50	35	.0264	.0265	.0081	**.0039**	FAIL	PASS	PASS	PASS
70	35	.0318	.0319	.0114	**.0049**	FAIL	PASS	PASS	PASS
70	45	.0379	.0382	**.0024**	.0061	FAIL	PASS	PASS	PASS
100	35	.0455	.0457	.0144	**.0061**	FAIL	PASS	PASS	PASS
100	50	.0559	.0559	**.0010**	.0061	FAIL	PASS	PASS	PASS

2) Key Findings: Five principal findings emerge from the seven scenarios:

a) (F1) GAPF consistently achieves the highest PDR, with the advantage growing in sparser networks.: GAPF's relative PDR improvement over the best standalone MARL baseline scales from $+35\%$ (n=20) to $+54\%$ (n=30), $+79\%$ (n=50), and peaks at $+111\%$ (n=100, R=35). Increasing the coverage radius narrows the gap (e.g., $+80\%$ at $(70, 35)$ vs $+47\%$ at $(70, 45)$; $+111\%$ at $(100, 35)$ vs $+43\%$ at $(100, 50)$), confirming that GAPF's topology-aware policy fusion is most valuable where routing is hardest—sparse networks with limited relay options.

b) (F2) GAPF achieves 2–$3.5\times$ *lower energy consumption and* 3–$9\times$ *better energy balance than QMIX/QTRAN.:* While QPSOFL uses the least total energy in most scenarios, its near-zero delivery renders this meaningless. Among algorithms with viable PDR ($> 50\%$), GAPF consistently uses the least energy. The energy balance ratio ($\sigma_E^{\text{QMIX}}/\sigma_E^{\text{GAPF}}$) widens from $1.7\times$ (n=20) to $9.2\times$ (n=100, R=50), confirming that GAPF's GCN-based load distribution scales with network size.

c) (F3) QMIX and QTRAN converge to statistically indistinguishable performance.: Across all scenarios, QMIX and QTRAN yield nearly identical PDR (within 1–2%), energy, and latency. This validates GAPF's design: the performance gain stems from the GCN-based topology-aware controller and policy fusion, not from expert selection alone.

d) (F4) QPSOFL collapses at scale; low energy reflects non-delivery.: QPSOFL's PDR degrades from 7.5% (n=20) to below 10% at n=100, R=35. Wider coverage radii partially recover delivery (44.8% at n=100, R=50), but performance remains unacceptable. Its excellent energy balance at high R (e.g., σ_E=0.001 at $(100, 50)$) is an artifact of minimal packet transmission rather than intelligent routing.

e) (F5) GAPF is the only algorithm passing all feasibility constraints.: QMIX/QTRAN **fail** the 1,024,MB memory constraint in every scenario (peak memory 3.2–13.8,GB), while GAPF stays within budget (684–951,MB). All algorithms pass the p_{99}<50,ms latency constraint. GAPF's memory footprint at n=100 (951,MB) leaves only 7% headroom, warranting attention for even larger deployments.

A. Deployment Feasibility on Resource-Constrained Nodes

The CTDE paradigm [49, 50] decouples training from deployment complexity. At inference time, the heavy training-only components—value-decomposition mixers, reward scalarization network, ES loop—are *discarded entirely*.

TABLE IX: Component deployment map under CTDE. Only the per-sensor agent and GAPF controller are active at inference.

Component	Runs on	Deployed?	Size
QMIX/QTRAN mixer, Q·K Attention, ES loop	Training server	No	—
GAPF meta-controller (GCN + CAS)	Sink/gateway	Yes (1×/episode)	∼1,473 params (∼6,KB)
Per-sensor agent (GRU)	Each sensor	Yes (1×/step)	∼25K params (∼100,KB)

Each sensor executes a single-layer GRU agent:

$$a_i = \arg\max_j; \mathbf{W}_2; \mathrm{GRU}!\big(\mathrm{ReLU}(\mathbf{W}_1\mathbf{o}_i + \mathbf{b}1), ; \mathbf{h}i^{(t-1)}\big) + \mathbf{b}_2 \tag{12}$$

where $\mathbf{o}_i \in \mathbb{R}^5$ is the local observation (remaining energy, consumed energy, x-y position, packet count), $\mathbf{h}_i^{(t-1)} \in \mathbb{R}^{64}$ is the recurrent hidden state, and $|\mathcal{A}|=N+1$. A single forward pass requires $\sim$29K MACs ($\sim$320 for the input layer, $\sim 24{,}576$ for the GRU cell, $\sim 4{,}544$ for the output layer at n=70), which completes in $<$ 1,ms even on an ARM Cortex-M4 at 80,MHz. The full model fits in $\sim$100,KB of flash, well within the capacity of commercial WSN platforms such as the STM32L4 (1,MB flash, 320,KB SRAM) or nRF52840 (1,MB flash, 256,KB RAM).

B. Limitations

Despite GAPF's strong empirical performance, several limitations warrant discussion:

1) **Simulation-only validation.** All experiments are conducted in a custom OpenAI Gym simulator. While we provide deployment feasibility analysis (Section IV-A) showing that the deployed agent requires only $\sim$100 lines of C and $< 30\,\mu$J per inference on a Cortex-M4 MCU, no hardware-in-the-loop experiments have been performed. Real-world factors such as channel fading, packet collisions, and clock drift are not modelled.
2) **Static topology assumption.** GAPF computes the graph embedding once per episode and selects a fixed expert for the entire episode. In scenarios with mobile sensors or rapidly changing channel conditions, step-level re-evaluation may be necessary. Three extensions can address this, ordered by complexity: (i) *periodic re-evaluation* every K steps with GRU hidden state reset (overhead: $\lceil T/K \rceil \times 0.03\,\text{ms}$); (ii) *confidence-triggered re-evaluation* when $|p_{\text{QMIX}} - 0.5| < \delta$; and (iii) *step-level Q-value fusion* blending both experts' outputs via a learned weight α_t at the cost of doubled per-step computation. In practice, WSN topologies change on the order of minutes to hours while a routing episode spans seconds, making option (i) sufficient for most deployments.
3) **Two-expert limitation.** The current CAS controller produces a binary selection (QMIX vs QTRAN). Extending to $k > 2$ experts (e.g., incorporating DQN, MAPPO, or QPLEX) would require a softmax-based selection mechanism and raises questions about diminishing returns as expert diversity increases.
4) **Fixed episode length.** All scenarios use $T = 100$ steps per episode. Real WSN deployments operate continuously with time-varying traffic patterns and gradual energy depletion over days or weeks. The episodic formulation does not capture long-horizon energy management.
5) **Memory constraint tightness.** At n=100, GAPF's memory footprint reaches 951 MB—within the 1 GB constraint but with only 7% headroom. Scaling beyond n=100 may require model compression techniques (pruning, quantisation) to remain feasible on resource-constrained gateways.
6) **Homogeneous sensor assumption.** All sensors are identical in hardware capability and

initial energy. Heterogeneous deployments with mixed sensor types, varying transmission power, or asymmetric energy budgets are not evaluated. However, the architecture accommodates heterogeneity with minimal changes: the node feature vector (Eq. 7) extends from $\mathbb{R}^4$ to $\mathbb{R}^{4+k}$ by appending sensor-specific attributes (battery capacity, transmission power, sensor type), requiring only a resized $\mathbf{W}_1 \in \mathbb{R}^{(4+k)\times d}$; the CAS controller and readout are unchanged. Heterogeneous radii yield an asymmetric adjacency handled by row-normalised propagation. For $k{=}4$ additional features the parameter increase is 64 (4% of 1,473), confirming no significant redesign is needed.

V. Conclusion

This paper introduced **Graph-Attentive Policy Fusion (GAPF)**, a lightweight meta-learning framework for energy-efficient multi-hop routing in Wireless Sensor Networks. GAPF fuses frozen MARL expert policies (QMIX, QTRAN) through a two-layer Graph Convolutional Network that encodes the sensor topology into a compact 16-dimensional embedding, enabling a Contextual Algorithm Selection controller to choose the best expert per deployment with only 1,473 trainable parameters.

A. Contributions and Impact

Extensive evaluation across 7 scenarios (20–100 sensors, 25–50 m coverage radius, 3 seeds, 21,000 test episodes) establishes three principal contributions:

1) **Near-perfect, scale-invariant delivery.** GAPF achieves $\geq$98% packet delivery ratio from 20 to 100 sensors, with +43% to +111% relative improvement over standalone MARL—critical for life-saving responsiveness in disaster monitoring.
2) **Energy-coherent routing for green AI.** GAPF consumes 2–3.5$\times$ less energy than QMIX/QTRAN while delivering more data, and maintains 3–9$\times$ better energy balance across sensors, directly extending network lifetime in battery-powered deployments.
3) **Feasible deployment on constrained hardware.** GAPF passes all feasibility constraints ($p_{99} < 50$ ms, peak memory $< 1{,}024$ MB) across all scenarios, while QMIX/QTRAN exceed the memory budget by up to $13\times$. Under CTDE, each sensor runs only a lightweight

GRU agent ($\sim$3 KB weights, $\sim$27 µJ per inference), compatible with commercial WSN microcontrollers.

B. Broader Impact

Beyond WSN routing, GAPF demonstrates that *lightweight policy fusion can outperform heavyweight monolithic models* in multi-agent systems. Encoding structural context via GNNs and selecting among frozen specialist policies is applicable to vehicular ad-hoc networks, drone swarm coordination, and smart grid load balancing. The meta-controller's small footprint (1,473 parameters) aligns with responsible AI: interpretable expert selection, minimal carbon footprint, and deployment without GPU infrastructure.

C. Future Work

We identify five directions for future research, ordered by expected impact:

1) **Hardware-in-the-loop validation.** Deploying GAPF on physical WSN testbeds (e.g., nRF52840 motes) with realistic channel models and packet collisions to bridge the gap between our deployment feasibility analysis and real RF conditions.
2) **Cross-domain disaster monitoring.** Adapting the framework to flood monitoring (mobile sensors), wildfire detection (rapidly evolving topology), and post-earthquake structural health monitoring (intermittent connectivity).
3) **Model compression for extreme scale.** Applying 8-bit quantisation, structured pruning, and knowledge distillation to extend feasibility beyond $n{=}100$ (where gateway memory reaches 951 MB).
4) **Dynamic expert fusion.** Extending the CAS controller to step-level Q-value blending to adapt within episodes to topology changes caused by sensor failures or energy depletion.
5) **Multi-expert generalisation.** Incorporating additional experts (QPLEX, MAPPO, DQN) with softmax-based selection to test whether expert diversity yields diminishing or compounding returns.

In summary, GAPF bridges the gap between the theoretical promise of MARL and the practical constraints of real-world sensor networks, achieving reliable, energy-efficient, and deployable routing at scale—a step toward responsible and sustainable AI for public safety.

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